USPTO Provisional Patent Application Cover Sheet (SB/16)

Title of Invention: Nose-Mounted Tactile Meditation Device

First Named Inventor: Michael Christopher Usowski

Residence (City & State): San Anselmo CA 94960

Mailing Address: 17 Parkside Court

Correspondence Address (Email/Phone): _____

Attorney/Agent (if applicable): N/A (filing pro se) Provisional

Application Type: Provisional Patent Application

Number of Drawing Sheets: 1

Number of Pages (including drawings): 6

Entity Status: ■ Micro Entity ☐ Small Entity ☐ Large Entity

Provisional Patent Application

Field of the Invention

This invention relates to the field of mindfulness, meditation, and psychotherapeutic tools, and more specifically to tactile devices that enhance breath awareness, concentration, and interoceptive presence by amplifying the felt experience of air moving across the skin. The device engages sensory neurons in the highly innervated area surrounding the nostrils and upper lip, supporting deeper embodiment and anchoring attention through consistent, clearly perceptible physical feedback.

Abstract

A nose-mounted, ergonomically shaped, meditation aid designed to amplify awareness of the in- and out-breath at the nostrils and upper lip (philtrum) by enhancing tactile and sensory perception in this concentrated neural region. The device mounts to the sides of the nostrils, with a main body positioned beneath them, to provide somatosensory engagement without obstructing airflow. By directing attention to the physical sensation of breathing, the device supports concentration, interoceptive awareness, and practices such as meditation, breathwork, and emotional regulation. Serving as a grounding anchor for breath awareness, the device assists users in maintaining present-moment focus and embodied calm, and may be particularly beneficial for individuals experiencing attentional drift, cognitive distraction, or emotional turbulence.

Background of the Invention

Existing meditation and mindfulness tools primarily rely on visual or auditory cues—such as timers, bells, apps, or guided audio—to help users maintain focus. Some wearable devices offer haptic feedback or sensor-based data collection but are often complex, tech-dependent, or limited in their ability to foster embodied presence. These tools may also divert attention outward rather than inward. There remains a clear gap in tools that offer direct, physical engagement with the breath at its most intimate and accessible point—the nostrils and upper lip (philtrum). Despite this area's high neural sensitivity and deep interoceptive potential, few if any meditation aids have been designed to amplify the tactile perception of airflow itself. This invention addresses that gap with a body-mounted device that delivers consistent tactile feedback precisely where breath is felt. In one embodiment, the device requires no external inputs or digital systems, offering a grounded, intuitive aid for attentional focus, somatic regulation, and meditative absorption. The result is a neurologically informed support tool that is portable, accessible, and of potential psychotherapeutic value.

Summary of Invention

The present invention is a nose-mounted tactile device configured to amplify physical awareness of the breath at its point of entry. In one form, the device includes a formable inner structure, such as a flexible metal wire or mesh, surrounded by a soft, skin-safe exterior material such as silicone or rubber. In another form, the device is manufactured as a molded soft body that retains its shape without internal reinforcement. In both cases, the device is shaped as a mirrored pair of S-curves that extend beneath and alongside the nostrils and continue upward along the sides of the nose. This configuration provides light tactile contact and a consistent cue during inhalation and exhalation. The device may be hand-molded or pre-formed to accommodate different nose shapes and sizes, and is designed to be comfortable, lightweight, and portable. Primary applications include meditation, breathwork, somatic regulation, trauma recovery, and daily calming practices. In certain embodiments, the device may also incorporate sensors, vibratory elements, or microelectronics to provide biofeedback, breath pacing guidance, or connectivity with external applications or data systems.

Detailed Description of the Invention

The present invention will be described with reference to the accompanying drawings, in which like reference numbers indicate corresponding parts. The embodiments described herein are illustrative and not limiting; variations in structure, material, and application remain within the scope of the invention.

Fig. 1 illustrates the device in a front view. The main body (10) is positioned directly beneath the nostrils, with curved arm sections (12) extending upward alongside the nose.

Fig. 2 shows a side view of the device. The main body (10) rests beneath the nostrils, while the curved arm sections (12) extend upward along the outer sides of the nostrils toward the bridge of the nose. A nasal engagement section (14) provides secure positioning and stability beneath and around the nostrils.

Fig. 3 shows a three-quarter perspective view. The device may include a flexible wire frame (16) or mesh core embedded within a soft encasing (18), such as silicone or rubber, allowing the user to hand-mold the device to fit their anatomy. The soft encasing (18) provides skin safety and comfort during extended use. Optional features may include a textured or extruded surface (20) located on the main body (10) or arm sections (12) to enhance tactile sensation of breath passing across the skin. An optional sensor module (22) may be embedded within the main body (10) or near the arm sections (12) to detect airflow, temperature, humidity, or biometric signals such as heart rate variability. The ergonomic curve of the device conforms to the user's nasal structure, ensuring both comfort and functional performance during meditation practice.

In one embodiment, the device includes an internal formable structure, such as wire (16) or mesh, surrounded by a soft exterior (18). This configuration allows the user to bend and shape the device for a secure fit on a wide range of nose sizes and shapes.

In another embodiment, the device is manufactured as a molded soft body without internal reinforcement. In this version, the inherent resilience of the molding material (e.g., silicone, rubber, or foam) provides sufficient conformity and stability to maintain contact with the nostrils.

In all embodiments, the device amplifies somatosensory perception by creating air acceleration and vortices around the point of breath entry, thereby enhancing awareness of airflow during inhalation and exhalation.

The device is designed to be comfortable, lightweight, and portable, supporting use in meditation, breathwork, somatic therapy, trauma recovery, and daily calming rituals. Optional features, such as sensor modules, micro-vibration elements, or wireless communication, may be incorporated either integrally or as modular attachments. These features can provide biofeedback, breath pacing cues, or data connectivity to external applications, extending the functionality of the device while remaining within the spirit and scope of the invention.

Alternate Embodiments

In addition to the primary forms described above, the invention may be adapted with variations in structure and functionality. In some embodiments, the device may be manufactured without an internal mesh or wire structure. In such cases, the device relies on the molding process and the inherent resilience of the chosen material (e.g., silicone, rubber, or foam) to conform to and remain in place on the nose.

The invention may also incorporate integrated or modular electronic components for enhanced functionality. These embodiments may include:

- Breath sensing elements such as airflow, temperature, or humidity sensors to detect inhale/exhale patterns.
- Micro-vibration modules to provide rhythmic cues or haptic pacing feedback.
- LED indicators or low-light visual cues embedded in the device for subtle breath reminders.
- Wireless communication modules (e.g., Bluetooth) for syncing with meditation apps or journaling platforms.
- Biometric feedback integration, such as heart rate variability (HRV) monitoring or stress indicators.

Claims

1. A nose-mounted tactile meditation device comprising:
   a) a body shaped as a mirrored pair of S-curves configured to extend beneath and alongside the nostrils and continue upward along the sides of the nose; and
   b) a skin-safe exterior material configured to provide tactile feedback at the point of breath entry during inhalation and exhalation.

2. The device of claim 1, wherein the body further comprises a formable inner structure selected from the group consisting of a flexible metal wire, mesh, or combination thereof, enclosed within the exterior material.

3. The device of claim 1, wherein the body is formed without an internal support structure and is manufactured as a molded soft body that retains its shape and conforms to the nose by resilience of the material.

Brief Description of Drawings

Fig. 1 shows the device in front view.
Fig. 2 shows the device in side view.
Fig. 3 shows the device in three-quarter perspective view.

Fig 1: Front View

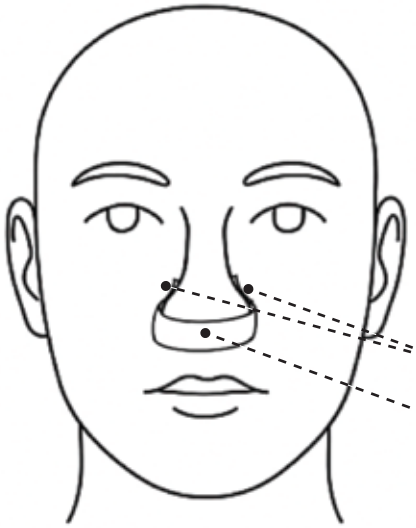

Fig 2: Side View

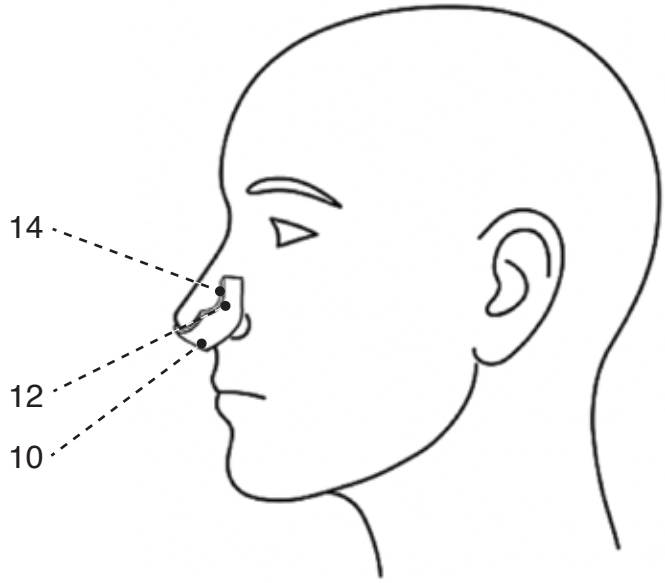

Fig 3: Three-Quarter Side View

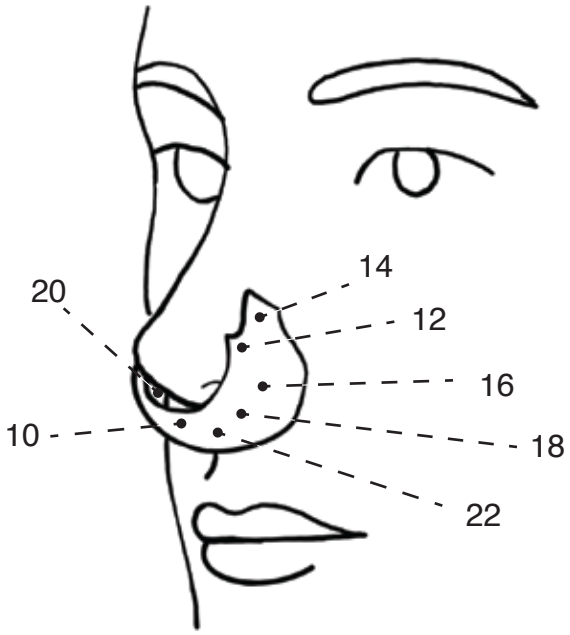

Figure 1-3 Legend and Labels

Main Body (10)
Portion that sits directly beneath the nostrils.

Curved Arm Sections (12)
“S-shaped” or curved arms that hug the nostrils and/or bridge of the nose.

Nasal Engagement Section (14)
The part that makes contact with or stabilizes under/around the nostrils.

Flexible Wire Frame (16)
The bendable frame or shaping element inside with an internal Wire / Mesh Core

Soft Encasing (18)
Silicone or other soft material that encases the wire frame.

Textured / Surface (20)
Raised ridges, soft pads, or textured tips for amplifying sensation and sensory feedback.

Optional Sensor Module (22)
If included, the location of embedded sensors (for breath, HR, vibration, etc.).